

BirdCLEF 2026: Bird Species Identification from Passive Acoustic Recordings

Data 266 Project Report

Team 7

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Abstract—Passive Acoustic Monitoring (PAM) has emerged as an important approach for large-scale biodiversity monitoring, enabling automated analysis of wildlife recordings collected from natural environments. The BirdCLEF 2026 challenge focuses on identifying bird species from passive acoustic recordings captured across Brazil’s Pantanal wetlands, where noisy soundscapes and overlapping vocalizations make classification difficult. In this project, we developed a deep learning pipeline for multi-label wildlife species classification using mel spectrogram representations of audio recordings. An EfficientNet-B2 model pretrained on ImageNet was used as the primary baseline architecture and fine-tuned on BirdCLEF training data. Multiple experiments were conducted to improve model robustness, including weighted binary cross-entropy loss for class imbalance handling, supervised soundscape fine-tuning for domain adaptation and limited Audio Spectrogram Transformer (AST) experimentation. Experimental results showed that models trained only on clean audio generalized poorly to real-world soundscape recordings despite strong clean validation performance. The most significant improvement was achieved through soundscape fine-tuning, which substantially improved soundscape validation performance and Kaggle leaderboard scores. Qualitative inference demonstrations further showed that the final model could successfully identify multiple bird species even under noisy acoustic conditions. The project highlights the importance of domain adaptation for real-world bioacoustic classification and demonstrates the effectiveness of spectrogram-based convolutional neural networks for large-scale bird species identification. In later stages of the project, a pretrained Perch v2 based inference pipeline was also evaluated and achieved the best public Kaggle leaderboard score of 0.889.

I. INTRODUCTION

PAM (Passive Acoustic Monitoring) emerged as an effective and efficient large scale monitoring technique for biodiversity because of its ability to capture wildlife sounds across continuous periods. With recent advances in deep learning and bioacoustic research in the last decade, species recognition from soundscape recordings is increasingly possible using machine learning techniques. Nevertheless, natural acoustic environments are still difficult due to overlapping calls, ambient noise, differences in audio recording, and imbalanced classes.

The BirdCLEF 2026 challenge, organized as part of the LifeCLEF evaluation initiative on Kaggle [1], focuses on automatic bird species identification from passive acoustic

recordings collected in Brazil’s Pantanal wetlands. The dataset contains both clean bird audio recordings and long-duration soundscape recordings annotated with multiple species labels. Participants are required to build robust multi-label classification systems capable of predicting bird species presence within short temporal windows of noisy environmental recordings.

In this project, we developed a deep learning pipeline for BirdCLEF 2026 using mel spectrogram representations generated from raw audio recordings. Our primary model architecture was EfficientNet-B2 pretrained on ImageNet and fine-tuned for multi-label bird species classification. Multiple experiments were conducted to improve generalization performance, including weighted binary cross-entropy loss for handling class imbalance, supervised soundscape fine-tuning for domain adaptation, and exploratory Audio Spectrogram Transformer (AST) experimentation. We also evaluated ensemble-based inference strategies that further improved leaderboard performance.

Experimental analysis showed that models trained only on clean bird recordings achieved high validation performance but generalized poorly to real-world soundscape recordings. The most significant improvements within the EfficientNet pipeline were obtained through supervised soundscape fine-tuning and ensemble inference. In later stages of the project, pretrained Perch v2 based bioacoustic inference pipelines were also evaluated and achieved the best Kaggle public leaderboard score of 0.889. The project demonstrates the importance of domain adaptation, pretrained bioacoustic foundation models, and robust inference strategies for practical wildlife classification systems operating in noisy real-world environments.

II. LITERATURE REVIEW AND RELATED WORK

Automatic bird species classification from environmental audio recordings has become an important research area in bioacoustics and biodiversity monitoring. Recent advances in deep learning have enabled machine learning systems to identify bird vocalizations directly from spectrogram representations of audio recordings. Convolutional neural networks (CNNs) are widely used for such tasks because spectrograms transform acoustic signals into image-like representa-

tions suitable for computer vision architectures. EfficientNet models proposed by Tan and Le [2] demonstrated strong performance and computational efficiency through compound model scaling. Due to their balance between accuracy and efficiency, EfficientNet architectures have been widely adopted for spectrogram-based audio classification tasks. In this project, EfficientNet-B2 was selected as the primary baseline architecture for bird species classification. BirdNET [3] introduced a deep learning framework specifically designed for large-scale avian diversity monitoring using passive acoustic recordings. The study demonstrated the practical usefulness of automated bird call detection systems for ecological monitoring and inspired many recent bioacoustic classification approaches. Mel spectrograms are commonly used in audio classification because they preserve important frequency characteristics of acoustic signals while reducing computational complexity. Previous studies on environmental sound classification and bird audio recognition have shown that mel spectrogram features combined with transfer learning can achieve strong classification performance [4]. Transformer-based audio architectures such as Audio Spectrogram Transformers (AST) have also shown promising performance in audio classification research [5]. Motivated by these developments, limited AST experimentation was explored during this project. However, due to runtime and computational constraints, the final competition pipeline primarily relied on EfficientNet-based convolutional models. Recent advances in large-scale pretrained bioacoustic foundation models have also improved wildlife audio understanding tasks. Google’s Perch model was developed for biodiversity monitoring using passive acoustic recordings and provides pretrained embeddings and classification capabilities for environmental sound analysis [6]. Motivated by the strong performance of such foundation models, Perch v2 based inference pipelines were also evaluated during the later stages of this project.

III. DATASET DESCRIPTION AND EXPLORATORY DATA ANALYSIS

A. Dataset Sources

The BirdCLEF+ 2026 data is stored on Kaggle and includes two types of recordings: one from the Xeno-canto community data collection and one from iNaturalist. Xeno-canto recordings are submissions by volunteer naturalists and generally consist of single-species recordings of high-quality audio with rich species and time metadata. iNaturalist recordings have additional species. These audio recordings were resampled to 32kHz and provided in OGG format as would be found in the test set. These are recordings from the Pantanal in Brazil, containing audio of birds, amphibians, mammals, reptiles, and insects. Along with the audio clips, the dataset includes additional metadata like species taxonomy, recording locations, recording quality, primary and secondary labels. All of this metadata came in handy when doing exploratory data analysis and when developing the models, especially when checking for class imbalances and multiple label cooccurrences.

B. Training Audio and Soundscape Data

The dataset comprises two distinct types of audio, each serving a different role in our pipeline.

Training Audio (`train_audio/`): This consists of 35,549 short labeled clips of individual species vocalizations, organized into subdirectories by species code. Clips are typically 5 to 30 seconds long and represent relatively clean, single-species recordings. These files form the primary supervised training set for all experiments.

Train Soundscapes (`train_soundscapes/`): These are long-duration continuous recordings made at Pantanal field sites using Autonomous Recording Units (ARUs), following the same naming convention as the hidden test soundscapes. Each file is approximately one minute long and was captured under realistic field conditions with ambient noise, overlapping species calls, and environmental interference. A labeled subset of 1,478 five-second segments from 66 unique soundscape files is provided in `train_soundscapes_labels.csv`, with columns specifying the filename, start time, end time, and a semicolon-separated list of species codes present in each segment.

Hidden Test Soundscapes (`test_soundscapes/`): Approximately 600 one-minute recordings are made available only during Kaggle notebook execution. These represent the actual evaluation set and cannot be accessed during local development.

C. Taxonomy and Label Structure

The competition defines 234 target species spanning five taxonomic classes: Aves (birds), Amphibia, Mammalia, Insecta, and Reptilia. Species labels follow eBird codes for birds and iNaturalist taxon IDs for non-bird species. Notably, some insect species are not identified at the species level and instead appear as sonotypes (e.g., 47158son01 through 47158son21). The taxonomy file provides mappings between primary labels, scientific names, common names, iNaturalist taxon IDs, and class names. The sample submission file defines the 234 species columns in the exact order required for evaluation. Key dataset statistics are summarized as follows: the training audio contains 35,549 clips across 206 unique primary labels; 234 species are required for submission; 1,478 labeled soundscape segments are available across 66 soundscape files.

D. Class Distribution and Imbalance

Exploratory analysis revealed a pronounced class imbalance across the 234 species. The most frequent classes contain several hundred recordings while the rarest species have only a single training example. This extreme long-tail distribution—where the ratio between the most and least represented species reaches several hundred times—poses a significant challenge for model training, as naive optimization tends to overfit to frequent species while neglecting rare ones. This severe imbalance motivated additional experiments using weighted BCE loss and soundscape fine-tuning during model development.

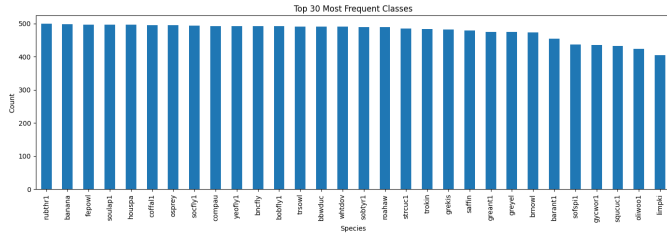


Fig. 1. Top 30 most frequently occurring species in the training dataset, ranked by clip count. A small number of species dominate the training distribution, while the majority of the 234 target classes have significantly fewer samples, illustrating the long-tail imbalance.

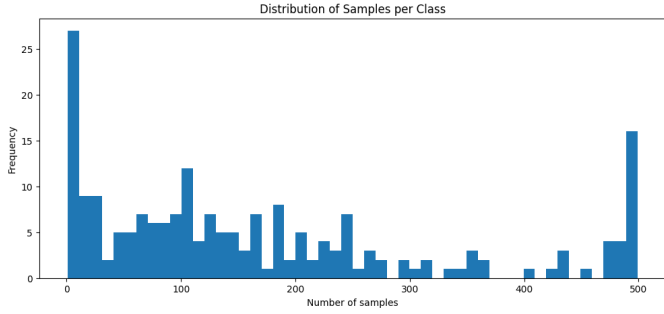


Fig. 2. Distribution of training samples per species class. The histogram reveals extreme class imbalance, with most species having fewer than 50 recordings.

E. Audio and Spectrogram Visualization

To understand the raw input data, we visualized audio waveforms and their corresponding mel spectrograms. Distinct call patterns are visible as bright horizontal or diagonal streaks against a darker background. Comparisons between training clips and soundscape segments visually confirm the domain gap: training spectrograms typically show a single dominant call pattern, while soundscape spectrograms display multiple overlapping patterns and a noisier background energy distribution.

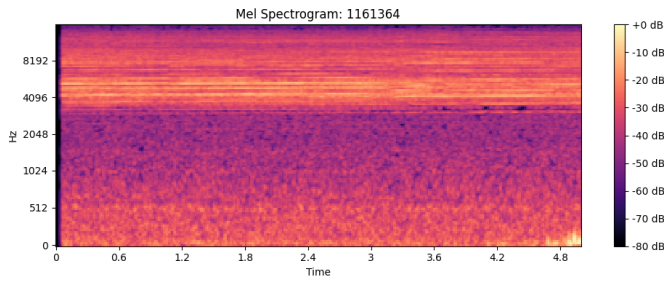


Fig. 3. Mel spectrogram of a representative training audio clip. The x-axis represents time and the y-axis represents mel-scaled frequency. Bright regions indicate high energy, corresponding to active vocalizations.

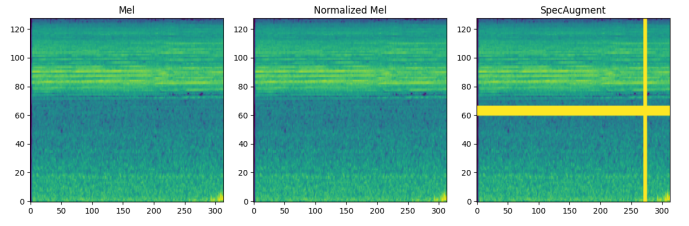


Fig. 4. Mel spectrogram transformations applied during preprocessing: (left) raw mel spectrogram, (center) normalized version, (right) after SpecAugment masking. SpecAugment randomly zeros out time and frequency bands to improve robustness.

IV. DATA PREPROCESSING AND FEATURE ENGINEERING

A. Audio Loading and Resampling

All audio files were loaded using the `librosa` library with a fixed target sample rate of 32 kHz, matching the resampling applied by the competition organizers to the test data. Loading was performed in mono mode to reduce computational cost and ensure consistency across recordings that may have been originally captured in stereo. Error handling was incorporated to return silent arrays for any files that failed to load, preventing training interruptions from corrupted files.

B. Five-Second Segmentation

This task presents the problem of classifying audio into a five second window task, where for every five second contiguous sequence of the test soundscape, the network needs to predict the likelihood of the occurrence of each species. In order to make sure that the training and inference process follow the same pattern, all training clips were trimmed to have exactly a five second duration. Shorter clips were zero-padded to make them five seconds long, while long clips were trimmed at the five second boundary.

C. Mel Spectrogram Generation

The raw audio waveform signals were transformed into mel spectrograms with optimal parameters for the best resolution in time and frequency for wildlife sounds. An FFT window of $n_fft=2048$ and hop length of 512, equal to 75% overlap between adjacent windows, was defined. Furthermore, there were 128 mel frequency bins (n_mels), while the frequency domain was reduced to 20–16,000 Hz.

In addition, the raw mel power spectrogram was transformed into dB using `librosa` library `librosa.power_to_db` function by setting the max parameter as the reference value. The reason for doing so is that the dB scale provides logarithmic compression, which enhances perceptually significant frequency changes and minimizes dynamic range.

D. Normalization, Resizing, and Channel Stacking

After transformation into decibel scale, the mel-spectrogram underwent normalization in the interval $[0, 1]$ through min-max normalization. The processed mel-spectrogram of dimension 128×313 was subsequently resized to 224×224 through

bilinear interpolation to meet the requirements of input dimensions in the EfficientNet-B2 backbone architecture. Lastly, the single-channel mel-spectrogram was replicated three times along the channel axis to result in the dimensionality of $3 \times 224 \times 224$, similar to an RGB image that the ImageNet-based CNN architecture requires.

E. Train-Validation Splitting

Stratified splitting was utilized for splitting the training labeled audios to have equal proportions of species in each set. Species that only had one training example were removed from the stratified split and kept in the training set alone to prevent splitting mistakes. An 80/20 split resulted in 28,436 training samples and 7,109 validation samples.

In soundscapes, stratified splitting using `GroupShuffleSplit` was employed on labeled soundscapes to ensure that soundscapes from the same file are kept together during training and testing to avoid data leakage problems. Stratified splitting generated 1,182 training soundscapes from 52 files and 296 validation soundscapes from 14 files.

V. PROBLEM FORMULATION AND MODEL SELECTION

A. Multi-Label Classification Formulation

The BirdCLEF 2026 task was formulated as a multi-label audio classification task, because more than one wildlife species could be vocally active in a single snippet. The model had to calculate the probability that any of the 234 target classes were present for each five-second audio window. Unlike typical single-label classification problems, an audio snippet may be labeled with multiple positive labels due to vocalizations from different species of birds, amphibians, insects, and mammals. Consequently, the model used sigmoid with binary cross-entropy based optimization, rather than softmax single-class based optimization.

B. EfficientNet-B2 Model Justification

EfficientNet-B2 was selected as the primary model architecture due to its strong balance between classification performance and computational efficiency. As mel spectrograms look more like images, it is reasonable to use pre-trained models such as EfficientNet from ImageNet as a transferable representation of a given image to classify an audio. Using EfficientNet-B2 for this project was convenient in many ways, specifically that it efficiently scaled up the parameters, had fairly smooth training dynamics, and had an input size compatible with 224×224 spectrogram inputs. The architecture also achieved strong validation performance during early baseline experiments, making it suitable for subsequent soundscape fine-tuning and leaderboard submissions.

C. Loss Function Selection

Binary Cross-Entropy (BCE) loss was used as the primary optimization objective because of the multi-label nature of the competition. Initial baseline experiments used standard BCE loss for independent class probability prediction. To address

severe class imbalance across species categories, additional experiments were conducted using weighted BCE loss. The weighted formulation assigned larger importance to under-represented classes during optimization. Although weighted BCE slightly improved Kaggle leaderboard performance, the most substantial gains were later achieved through soundscape domain adaptation.

D. Soundscape Domain Adaptation Motivation

One of the major challenges in the competition was the domain gap between clean training recordings and noisy real-world soundscapes. Models trained only on isolated bird recordings achieved strong clean validation performance but generalized poorly to environmental soundscape recordings containing overlapping calls, background insects, wind noise, and other environmental artifacts. To address this issue, supervised soundscape fine-tuning was performed using labeled five-second soundscape segments. This domain adaptation strategy allowed the model to learn acoustic patterns more representative of the hidden competition test environment and resulted in significant improvements in soundscape validation and Kaggle leaderboard performance.

E. Audio Spectrogram Transformer (AST) Experimentation

In addition to convolutional neural networks, limited experimentation was conducted using Audio Spectrogram Transformer (AST) architectures for spectrogram-based audio classification. Transformer-based models have recently shown promising performance in audio understanding tasks due to their ability to capture long-range temporal relationships within spectrogram representations. However, due to the runtime limit imposed on the Kaggle notebook and the computational cost of transformer-based inference, we were not able to run AST on larger scales. As a result, the final competition submissions primarily relied on EfficientNet-based pipelines.

F. Alternative Models Considered

In addition to the EfficientNet-B2 training pipeline, pre-trained bioacoustic foundation models were also explored during the later stages of the project. In particular, the Perch v2 CPU model released for BirdCLEF 2026 was evaluated using probe-based inference on soundscape recordings. Unlike the EfficientNet experiments, which required supervised spectrogram training and soundscape fine-tuning, the Perch model provided pretrained bioacoustic embeddings and species prediction capabilities specifically designed for wildlife acoustic monitoring tasks. The Perch-based inference pipeline achieved the best Kaggle leaderboard performance of the project with a public leaderboard score of 0.889. Although EfficientNet-based experiments formed the primary development and experimentation framework throughout the project, the final leaderboard results demonstrated the effectiveness of large-scale pretrained bioacoustic foundation models for real-world passive acoustic monitoring tasks.

VI. METHODOLOGY AND EXPERIMENTAL SETUP

A. Baseline EfficientNet-B2 Model

The primary baseline model used in this project was EfficientNet-B2 pretrained on the ImageNet dataset. Mel spectrogram representations generated from five-second audio segments were used as model inputs after resizing to 224×224 resolution and converting to three-channel tensors. The baseline training pipeline was implemented using PyTorch and optimized using Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss) for multi-label classification. Initial experiments were performed on clean training recordings before extending the pipeline to soundscape adaptation experiments. The baseline EfficientNet-B2 model achieved strong clean validation performance with a best validation AUC of approximately 0.9576 during experimentation. Training procedures included mini-batch optimization, validation AUC monitoring, checkpoint saving, and experiment tracking across multiple epochs. The EfficientNet pipeline formed the primary experimentation framework throughout the project.

B. Weighted BCE Experiment

One major challenge in the BirdCLEF dataset was severe class imbalance across species categories. Some species had thousands of training examples, while others had only limited recordings. To address this issue, an additional experiment was conducted using a weighted Binary Cross-Entropy loss function. The weighted BCE experiment assigned larger penalties to underrepresented classes during optimization in order to encourage improved minority-class learning. Experimental comparisons showed that weighted BCE slightly improved Kaggle leaderboard performance compared to the baseline configuration, although validation AUC remained similar to the original BCE setup. This experiment demonstrated that loss weighting alone was insufficient to fully solve the domain mismatch problem between clean recordings and real-world soundscapes.

C. Soundscape Fine-Tuning

The most significant performance improvement during the project was achieved through supervised soundscape fine-tuning. Although the baseline EfficientNet model performed well on the validation set of clean data, it generalised badly to an environment that contained multiple, overlapping sounds such as bird calls, noise from insects and wind. To address this issue, labeled five-second soundscape segments from `train_soundscape_labels.csv` were used for additional supervised training. Group-based train-validation splitting was applied at the soundscape filename level to avoid leakage between training and validation recordings. The soundscape fine-tuning process allowed the model to learn acoustic patterns more representative of the hidden competition test environment. This domain adaptation strategy substantially improved soundscape validation performance and Kaggle leaderboard scores. The final EfficientNet soundscape fine-tuned inference pipeline achieved a public leaderboard score of approximately 0.813.

D. AST Experimental Pipeline

Limited experimentation was also conducted using Audio Spectrogram Transformer (AST) architectures for spectrogram-based audio classification. AST models process spectrogram inputs using transformer-based attention mechanisms instead of convolutional feature extraction. The AST pipeline used spectrogram-based feature extraction together with transformer-based audio classification models implemented using PyTorch and Hugging Face Transformers. Initial AST experiments showed encouraging validation performance during training, although the models were not used for final leaderboard submissions due to runtime and computational constraints. Compared to EfficientNet-based pipelines, AST models required significantly higher computational resources and longer inference times. Due to Kaggle notebook runtime limitations and transformer inference overhead, AST experimentation could not be evaluated at larger scale.

E. Perch v2 Bioacoustic Foundation Model

During the later stages of the project, Google's pretrained Perch v2 CPU model was evaluated for wildlife soundscape classification. Unlike the EfficientNet-based experiments, which relied on supervised spectrogram training, Perch provided pretrained bioacoustic embeddings and species prediction capabilities specifically designed for passive acoustic monitoring tasks. The Perch inference pipeline processed one-minute soundscape recordings by dividing them into twelve five-second audio windows. Each segment was passed through the pretrained Perch model to generate embeddings and species prediction scores. Additional probe-based inference and prior fusion strategies were used to adapt the pretrained outputs to the BirdCLEF 2026 taxonomy and soundscape environment. The Perch v2 probe-based inference pipeline achieved the best public Kaggle leaderboard score of 0.889 during the project. The results demonstrated the effectiveness of large-scale pretrained bioacoustic foundation models for real-world wildlife monitoring applications.

F. Inference Pipeline

The final inference pipeline processed one-minute soundscape recordings by dividing them into twelve non-overlapping five-second windows. Mel spectrograms were generated for each segment and passed through trained classification models to obtain species probability predictions. Multiple inference strategies were evaluated during the project, including EfficientNet-based soundscape inference, ensemble-based inference, and pretrained Perch v2 probe-based inference pipelines. The EfficientNet inference system used fine-tuned soundscape checkpoints for prediction generation on environmental recordings. Although EfficientNet-based experiments formed the primary development and experimentation framework throughout the project, the final leaderboard results demonstrated the effectiveness of large-scale pretrained bioacoustic foundation models for real-world passive acoustic monitoring tasks.

G. Tools and Computational Environment

The project was implemented primarily using Python, PyTorch, TensorFlow, and Hugging Face Transformers. Audio preprocessing and spectrogram generation were performed using Librosa, NumPy, Pandas, and SoundFile utilities. Model training and inference experiments were conducted using Kaggle Notebooks with GPU acceleration where applicable. EfficientNet-B2 models were implemented using PyTorch-based deep learning workflows, while AST experimentation utilized transformer-based audio classification pipelines. The Perch v2 inference experiments were performed using TensorFlow-based pretrained bioacoustic models provided through the BirdCLEF competition resources. Experiment tracking included validation AUC monitoring, checkpoint saving, leaderboard submission tracking, and comparison of multiple model configurations across baseline, weighted BCE, soundscape fine-tuning, AST, ensemble, and Perch-based inference experiments.

VII. EVALUATION, RESULTS, AND ANALYSIS

A. Evaluation Metrics

Two primary metrics were used throughout this project. **Macro ROC-AUC** served as the main development metric during all training runs. For each species class, the ROC-AUC score measures how well the model ranks positive examples above negative ones, regardless of prediction threshold. Classes with no positive examples in the validation set were excluded from the macro average to avoid undefined values. This metric was chosen for monitoring because it provides a stable, threshold-independent signal of model quality across all 234 species simultaneously.

The **Kaggle public leaderboard score** uses padded class-mean Average Precision (cmAP) as its official metric. Before computing per-class AP, five artificial positive rows are prepended for each species. This padding prevents species with zero true positives in the hidden test set from producing undefined AP values and provides a more stable evaluation signal. Although both metrics generally agree in direction, the Kaggle cmAP tends to reward models that make confident, well-calibrated predictions on rare and infrequent species.

B. Experiment Tracking

The training histories for each of the conducted experiments were recorded in CSV files that were stored in a structured folder on Google Drive dedicated to this particular project. As a result, comparison between loss trajectories and AUC dynamics for each of the runs could be made without any additional execution of the experiment.

Baseline EfficientNet-B2 was trained for 10 epochs. Training loss steadily declined from 0.0307 at epoch 1 to 0.0025 at epoch 10. On the other hand, the validation loss was quickly reduced during the first three epochs and then stabilized at 0.013 for the remaining five training epochs. The absence of any overfitting during the first part of training shows good convergence in learning processes for the model. At the same time, validation AUC steadily increased until it hit 0.9576

at epoch 6. After that point, the validation AUC started to oscillate between 0.9526 and 0.9566 and did not increase further. These results suggest that the baseline model had largely saturated its performance on the clean audio validation set.

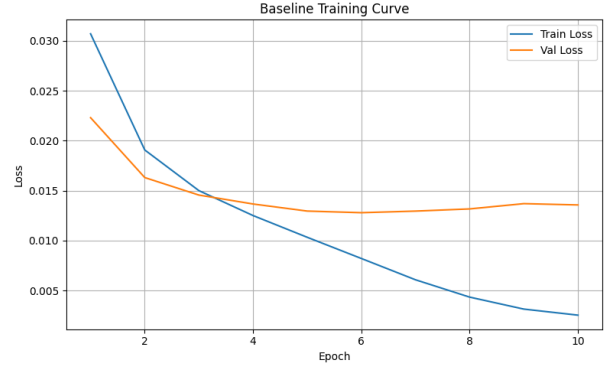


Fig. 5. Baseline EfficientNet-B2 training and validation loss curves over 10 epochs. Both losses decrease consistently, confirming stable convergence without severe overfitting.

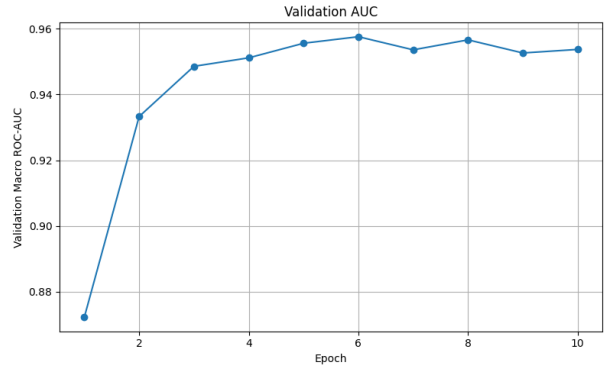


Fig. 6. Validation macro ROC-AUC for the baseline EfficientNet-B2 model across 10 epochs. Peak performance of 0.9576 was achieved at epoch 6.

C. Experiment Summary Table

All experiment configurations, results, and training histories were logged to structured CSV files and organized in a project directory on Google Drive. A consolidated experiment tracking table compares all four primary experiments side by side, allowing direct comparison of validation AUC across conditions.

Table I consolidates the configurations and results across all six experiments. Experiments E1 through E6 were conducted on EfficientNet-B2 and AST. The Perch v2 + linear probe pipeline (E7) used pretrained bioacoustic embeddings followed by lightweight logistic regression classifiers and is discussed separately due to its fundamentally different training paradigm. In addition to validation AUC, multiple Kaggle leaderboard submissions were evaluated to measure real-world soundscape generalization performance. The best leaderboard score achieved in this project was **0.889** using the Perch v2 +

linear probe pipeline, while the best supervised EfficientNet-based ensemble achieved **0.813**.

TABLE I
FULL EXPERIMENT SUMMARY — ALL RESULTS

Model	Training Data	Val AUC
EfficientNet-B2 (baseline)	train_audio	0.9576
EfficientNet-B2 (weighted BCE)	train_audio	0.9559
E1 checkpoint (no fine-tune)	Eval on soundscapes	0.6487
EfficientNet-B2 (soundscape FT)	train_soundscapes	0.8428
AST (clean)	train_audio	0.9500
AST (soundscape FT)	train_soundscapes	0.8135
Perch v2 + Linear Probes	train_soundscapes	0.7433 (val)

D. Clean Validation Results

The clean validation experiments demonstrated that EfficientNet-B2 was highly effective for supervised classification of curated single-species recordings. The baseline EfficientNet-B2 model (E1) achieved the best overall clean validation performance, reaching a peak macro ROC-AUC of 0.9576 after 10 epochs of training. Validation performance improved rapidly during the first few epochs before stabilizing near convergence, indicating that the model was able to learn discriminative acoustic patterns efficiently from mel spectrogram representations.

The weighted BCE variant (E2) produced very similar clean validation performance, achieving a peak AUC of 0.9559. Although the final AUC was slightly lower than the baseline configuration, weighted BCE demonstrated faster early convergence and later contributed to stronger leaderboard performance when combined with soundscape adaptation and ensemble inference. This suggests that class reweighting alone does not significantly improve clean-domain validation accuracy, but may still improve calibration and robustness for rare species during real-world inference.

The AST-based model (E5) also achieved strong clean-domain performance, reaching a validation AUC of 0.9500 after 6 epochs. However, transformer-based training introduced substantially higher computational overhead and slower convergence during the early epochs compared to EfficientNet-B2. Despite these limitations, AST remained competitive on clean validation data and demonstrated that attention-based architectures are capable of learning meaningful acoustic representations for large-scale bioacoustic classification tasks.

Overall, all clean-audio experiments achieved high validation AUC scores above 0.95, confirming that mel spectrogram representations combined with transfer learning provide highly effective performance on curated wildlife recordings. However, later soundscape experiments revealed that strong clean validation performance alone does not guarantee robust generalization to noisy real-world environmental recordings. While EfficientNet-B2 and AST were trained directly on mel spectrogram inputs, an additional embedding-based approach using pretrained Perch v2 representations and linear probes was also explored separately. Because Perch operates as a

pretrained bioacoustic foundation model rather than a fully trainable end-to-end classifier, its results are discussed independently in the subsequent discussion and leaderboard analysis sections.

E. Soundscape Validation Results

The most important findings of the project emerged from evaluation on the labeled soundscape validation dataset. Although the baseline EfficientNet-B2 model achieved excellent performance on clean validation recordings, its performance dropped substantially when evaluated directly on noisy environmental soundscapes. The baseline model achieved only 0.6487 AUC on soundscape validation data compared to 0.9576 on the clean validation set, revealing a severe domain gap between curated single-species recordings and real-world environmental audio. Supervised soundscape fine-tuning (E4) significantly improved soundscape performance. Fine-tuning the EfficientNet-B2 model on 1,182 labeled soundscape segments produced consistent AUC improvements across all epochs and achieved a final soundscape validation AUC of 0.8428. This represents an improvement of 0.1941 AUC points compared to the baseline model evaluated directly on soundscape recordings. These results demonstrate that even a relatively small amount of domain-matched training data can substantially improve generalization to realistic passive acoustic monitoring environments. The AST soundscape fine-tuning experiment (E6) demonstrated a similar trend. After fine-tuning on labeled soundscape segments, AST achieved a peak soundscape validation AUC of 0.8135. Although the final AST performance remained below the EfficientNet-B2 soundscape fine-tuned model, the experiment confirmed that domain adaptation consistently improves real-world soundscape performance across multiple architectures.

F. Perch v2 with Linear Probes (E7)

As an additional approach introduced after the initial set of experiments, we incorporated Google’s Perch v2 bird vocalization classifier through a probe-based inference strategy. Perch is a TensorFlow SavedModel trained by Google on a large corpus of bird vocalizations with learned embeddings optimized for acoustic species identification. Unlike EfficientNet and AST, Perch was used purely as a feature extractor rather than a classifier.

The pipeline operated as follows. First, 1536-dimensional embeddings were extracted from all labeled train soundscape windows using Perch’s `serving_default` inference signature, which processes 5-second audio chunks and outputs both classification logits over its internal species taxonomy and dense embedding vectors. These embeddings and their raw logit scores were cached as `full_perch_arrays.npz` to avoid recomputation. Second, 1280-dimensional embeddings were extracted from the full 35,705 training audio clips and cached as `train_embeddings.npz`.

The embedding dimensionality was reduced to 64 components using PCA with standard scaling applied beforehand, fitted separately for the soundscape embeddings (1536-d)

and the train audio embeddings (1280-d). Logistic regression probes with regularization strength $C = 0.25$ and balanced class weighting were then trained on the soundscape PCA embeddings for all 63 species classes that had at least 3 positive examples in the 708 fully-labeled soundscape windows. During test inference, the final prediction for each window was blended as:

$$s_c^{\text{final}} = (1 - \alpha) \cdot s_c^{\text{Perch}} + \alpha \cdot d_c^{\text{probe}} \quad (1)$$

where s_c^{Perch} is the raw Perch logit for class c , d_c^{probe} is the logistic regression decision function score, and $\alpha = 0.35$ is the probe blending weight. Additional post-processing steps were applied: unmapped inactive classes (species not appearing in the labeled soundscape data and not covered by Perch’s taxonomy) were assigned a suppression score of -8.0 ; temporal smoothing with $\alpha = 0.15$ was applied across consecutive windows for textured taxa (Amphibia, Insecta); and event smoothing with $\alpha = 0.10$ was applied for discrete call events. A temperature of 1.15 was applied before the final sigmoid to calibrate prediction confidence.

Out of the 234 competition species, 203 were directly mapped to Perch’s internal taxonomy through scientific name matching. The remaining 31 unmapped species included primarily insect sonotypes and some amphibian taxa not represented in Perch’s label set; for these, proxy genus-level matches were computed where possible, particularly for 3 amphibian classes.

G. Effect of Domain Adaptation

The results from all experiments clearly showed the importance of domain adaptation. The baseline EfficientNet-B2 model achieved 0.9576 AUC on the clean validation set, but its performance dropped to 0.6487 when evaluated directly on soundscape recordings. After soundscape fine-tuning, the same model improved to 0.8428 AUC on the soundscape validation set, giving an improvement of 0.1941 AUC points. A similar trend was observed for AST models, where soundscape fine-tuning improved the final soundscape AUC to 0.8135. These results indicate that exposure to realistic environmental recordings during training is more important for soundscape performance than changing the loss function or model architecture alone.

H. Result Analysis

Several important insights can be derived from the overall experimental results.

Pretrained Perch embeddings were highly effective for this task. The Perch v2 probe-based inference pipeline achieved the best public Kaggle leaderboard score of 0.889, substantially outperforming the best EfficientNet-based ensemble score of approximately 0.813. This result demonstrates the effectiveness of large-scale domain-specific pretraining for wildlife acoustic monitoring tasks. The pretrained Perch representations captured acoustic patterns that were difficult to recover through supervised fine-tuning of general-purpose image classification architectures alone.

Soundscape fine-tuning was the most impactful supervised improvement. Among all gradient-based experiments, supervised soundscape adaptation produced the largest performance gain. The EfficientNet-B2 baseline improved from 0.6487 AUC to 0.8428 AUC on soundscape validation data after fine-tuning on labeled soundscape segments. This improvement confirmed that exposure to realistic environmental recordings was critical for reducing the domain gap between clean curated recordings and noisy real-world soundscapes.

Weighted BCE provided only limited gains. Although weighted BCE slightly improved leaderboard performance and demonstrated faster early convergence during training, the final clean validation AUC remained very similar to the standard BCE baseline. These results suggest that class reweighting alone was insufficient to address the broader challenges of severe class imbalance and domain mismatch present in the BirdCLEF dataset.

Model diversity improved ensemble performance. Ensemble inference using multiple training strategies improved leaderboard performance compared to individual EfficientNet configurations. Combining weighted BCE and soundscape fine-tuned EfficientNet models achieved stronger generalization on hidden soundscape recordings, indicating that different optimization strategies captured complementary acoustic information useful for robust environmental audio classification.

VIII. VISUALIZATION AND QUALITATIVE INFERENCE DEMONSTRATION

A. Mel Spectrogram Examples

To investigate the structure of the acoustics of the soundscape recordings, mel spectrograms are plotted using the same pre-processor as in the training, but on a five-second sample from the validation set. The spectrograms are generated using the `librosa.display.specshow` with mel-scaled frequency axes and decibel-scale intensity mapping.

Figure 7 depicts a portion of a soundscape featuring many individual sources of vocalizations. Multiple harmonic patterns and bursts can be seen in various regions of the frequencies.

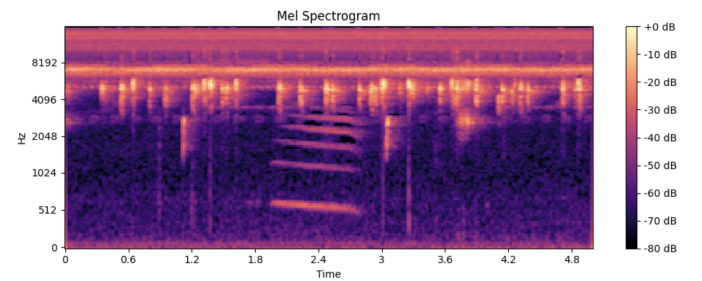


Fig. 7. Mel spectrogram of a validation soundscape segment containing multiple localized vocalization patterns

The validation segment in Figure 8 depicts a more challenging soundscape for a wildlife classifier, as the spectrogram illustrates a denser acoustic environment that is cluttered with background sounds and overlapping spectral signatures.

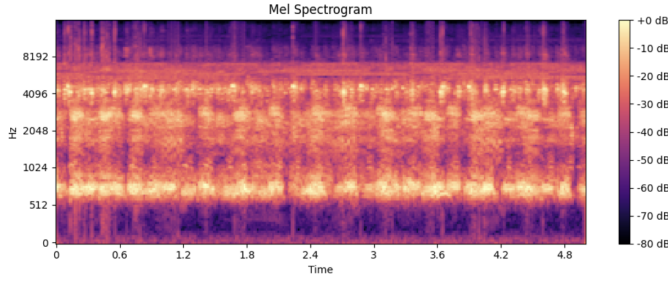


Fig. 8. Mel spectrogram of a noisy soundscape validation segment with overlapping acoustic activity and dense environmental background noise.

B. Soundscape Inference Demo

A qualitative inference demonstration was performed using held-out validation segments from the labeled soundscape dataset. As Kaggle’s hidden test recordings are unavailable outside the context of Kaggle’s notebook execution, inference analysis was performed on soundscape samples from the Kaggle validation dataset. These were drawn from real-world environmental recordings.

The trained soundscape fine-tuned EfficientNet-B2 checkpoint (`effb2_soundscape_finnetuned.pth`) was loaded in evaluation mode. Each five-second audio segment was converted into a mel spectrogram, normalized, resized to 224×224 , stacked into three channels, and passed to the model as a tensor of shape $1 \times 3 \times 224 \times 224$. Final species probabilities were generated by applying the sigmoid activation function to the model logits.

Loaded model from: `/content/drive/MyDrive/BirdCLEF_2026_Team7/checkpoints/effb2_soundscape_finnetuned.pth`

Fig. 9. Loading the soundscape fine-tuned EfficientNet-B2 checkpoint for qualitative inference demonstration on validation soundscape segments.

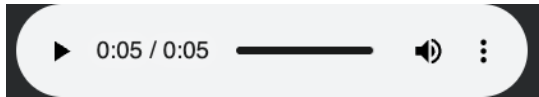


Fig. 10. Five-second validation soundscape audio segment used for qualitative inference analysis. Audio playback was used to manually inspect overlapping acoustic events and environmental noise conditions.

Model input shape: `torch.Size([1, 3, 224, 224])`

Fig. 11. Inference preprocessing output showing the final tensor shape $1 \times 3 \times 224 \times 224$ used as input to the EfficientNet-B2 model.

C. Top-K Prediction Analysis

To further analyze qualitative model behavior, the top predicted species probabilities were examined for multiple validation soundscape segments. For each five-second recording window, the model generated probability scores for all target species, and the top-ranked predictions were extracted using descending probability ordering.

Figure 12 shows the top-10 predictions for the first validation sample. The soundscape fine-tuned EfficientNet-B2 model successfully identified multiple ground-truth species within its highest-confidence predictions despite the presence of overlapping environmental noise and competing acoustic activity.

	rank	species	probability
0	1	65380	0.906181
1	2	517063	0.373404
2	3	23158	0.235624
3	4	22973	0.082190
4	5	24321	0.046231
5	6	fotfly	0.021634
6	7	trsowl	0.013657
7	8	47144	0.010445
8	9	555146	0.010237
9	10	66971	0.009724

True labels: `['22961', '23158', '65377', '65380']`

Fig. 12. Top-10 predicted species probabilities for the first validation soundscape segment. Several ground-truth species appear among the highest-confidence predictions generated by the model.

Figure 13 presents predictions for a second validation soundscape segment containing multiple simultaneously active species. Even under noisy acoustic conditions, the model successfully retrieved all ground-truth species within its top-ranked predictions, demonstrating strong robustness to real-world soundscape complexity.

	rank	species	probability
0	1	65380	0.985859
1	2	555146	0.973111
2	3	24321	0.903461
3	4	66971	0.829795
4	5	517063	0.125931
5	6	24279	0.058529
6	7	undtin1	0.006876
7	8	22973	0.005379
8	9	sibtan2	0.003126
9	10	23158	0.002553

True labels: `['24279', '555146', '65380', '66971']`

Fig. 13. Top-10 predicted species probabilities for a second soundscape validation segment. The model successfully recovered all ground-truth labels within its top predictions despite dense background acoustic activity.

These qualitative inference demonstrations support the quantitative soundscape validation results and illustrate the

effectiveness of supervised soundscape fine-tuning for improving real-world acoustic scene understanding.

IX. DISCUSSION

A. What Worked

Perch v2 embeddings with linear probes delivered the strongest overall performance. Using Perch purely as a fixed feature extractor and fitting lightweight logistic regression probes on the resulting soundscape-domain embeddings proved more effective than any gradient-trained model in our pipeline. The key advantage is that Perch’s representations were learned on a large corpus of bird vocalizations using contrastive objectives optimized for acoustic discriminability. The Perch v2 probe-based inference pipeline ultimately achieved the best public Kaggle leaderboard score of 0.889 during the project.

Soundscape fine-tuning was the most impactful improvement within the EfficientNet-based training pipeline. The increase from 0.6487 AUC to 0.8428 AUC on soundscape validation data demonstrated that even a relatively small amount of domain-matched labeled data could substantially reduce the domain gap between clean training recordings and real-world environmental soundscapes. Similar trends were also observed during AST experimentation, suggesting that the benefits of domain adaptation were not architecture-specific.

EfficientNet-B2 Ensemble inference also improved leaderboard performance compared to individual EfficientNet configurations. Combining predictions from weighted BCE and soundscape fine-tuned EfficientNet-B2 models produced stronger generalization on hidden test soundscapes and achieved a public Kaggle leaderboard score of approximately 0.813. This result suggests that different training objectives and fine-tuning strategies captured complementary acoustic information, allowing ensemble inference to improve robustness on noisy environmental recordings.

Spectrogram-based preprocessing and augmentation also contributed positively to model robustness. Mel spectrogram conversion allowed audio signals to be represented as image-like inputs suitable for convolutional architectures, while normalization and SpecAugment-style masking improved robustness to variability in environmental recordings. These preprocessing steps helped the model generalize better to noisy soundscape conditions containing overlapping species calls and background acoustic activity.

B. What Did Not Work

Weighted BCE loss provided only limited improvement compared to the standard BCE baseline. The weighted loss formulation was introduced to address severe class imbalance by assigning larger penalties to underrepresented species during optimization. Although weighted BCE slightly improved Kaggle leaderboard performance, the clean validation AUC remained very similar to the standard BCE configuration. These results suggest that class reweighting alone was not sufficient to address the broader domain mismatch and data scarcity

challenges present in the BirdCLEF dataset. In comparison, supervised soundscape fine-tuning produced substantially larger gains in real-world soundscape performance.

Standard SpecAugment alone could not bridge the domain gap between clean recordings and real-world soundscapes. Although SpecAugment improved robustness within the clean audio domain, it could not fully replicate the environmental characteristics of Pantanal field recordings, including persistent background insects, wind noise, overlapping vocalizations, and complex acoustic interference patterns. The substantial drop from clean validation performance to soundscape validation performance demonstrated that synthetic augmentation alone was not an adequate substitute for domain-matched soundscape training data.

C. AST Runtime and Computational Constraints

The Audio Spectrogram Transformer (AST) pipeline introduced considerably higher computational overhead compared to EfficientNet-based models. Transformer-based spectrogram processing required substantially longer training and inference times due to the self-attention operations applied over spectrogram token sequences. While AST experiments showed promising validation performance, large-scale experimentation was constrained by Kaggle notebook runtime limits, GPU memory requirements, and slower inference throughput. Compared to EfficientNet-B2, AST models were significantly more resource-intensive and required careful batch-size management during training and inference to avoid runtime failures.

D. Challenges and Limitations

Hidden test set: The competition’s code submission format meant that no local evaluation on actual test data was possible. All development decisions were made based on soundscape validation AUC as a proxy for leaderboard performance, which may not perfectly reflect the test distribution.

Limited labeled soundscape data: Only 1,478 segments with ground-truth expert annotations were available for soundscape fine-tuning. Expanding this to the full unlabeled soundscape archive through semi-supervised or self-supervised methods could substantially improve domain adaptation without requiring additional annotation effort.

Class imbalance: Despite weighted BCE and secondary label inclusion, models continued to struggle on rare species with very few training examples. Some species have only a single labeled training clip, making any form of supervised learning extremely challenging for those classes.

Non-bird species: This competition includes amphibians, reptiles, mammals, and insects in addition to birds. Most prior work on acoustic classification focuses exclusively on bird vocalizations, and pretrained models designed for bird identification provide limited utility for the full multi-taxon label set.

Computational and foundation model constraints: Although transformer-based AST models and the pretrained Perch v2 bioacoustic foundation model showed strong potential during experimentation, both approaches introduced

additional computational and practical challenges. AST architectures required substantially higher computational resources, GPU memory, and inference time compared to EfficientNet-based pipelines, limiting large-scale experimentation under Kaggle runtime constraints. Similarly, while the Perch v2 pipeline achieved the best leaderboard performance, it relied heavily on pretrained external representations with more limited interpretability and reduced control over task-specific feature learning compared to the custom EfficientNet training pipeline developed during this project.

X. CONCLUSION

In this project, we developed and evaluated multiple deep learning pipelines for the BirdCLEF 2026 wildlife acoustic classification challenge using passive acoustic monitoring recordings collected from the Pantanal wetlands of Brazil. The project explored spectrogram-based convolutional neural networks, transformer-based audio architectures, soundscape domain adaptation strategies, and pretrained bioacoustic foundation models for multi-label wildlife species classification.

The EfficientNet-B2 baseline achieved strong clean validation performance, demonstrating the effectiveness of mel spectrogram representations combined with transfer learning from ImageNet-pretrained convolutional architectures. However, experimental analysis showed that models trained only on clean audio recordings generalized poorly to noisy environmental soundscapes, highlighting the significant domain gap present in real-world passive acoustic monitoring tasks. Supervised soundscape fine-tuning substantially improved soundscape validation performance and leaderboard results, while ensemble inference across weighted BCE and soundscape fine-tuned EfficientNet models achieved a public Kaggle leaderboard score of approximately 0.813. These results confirmed the importance of domain adaptation and robust inference strategies for environmental audio classification systems.

Additional experiments using weighted BCE loss, AST architectures, ensemble inference, and Perch v2 bioacoustic foundation models provided further insights into the strengths and limitations of different modeling strategies. While AST models showed promising performance, their computational cost limited large-scale experimentation under Kaggle runtime constraints. In contrast, pretrained Perch v2 embeddings combined with lightweight probe-based inference achieved the strongest overall performance, producing the best public Kaggle leaderboard score of 0.889 during the project.

Our results indicate that successful classification of wildlife calls needs not only solid modelling approaches, but also attention paid to factors such as domain mismatch, noisy background sounds, class imbalance, and realistic soundscapes in the field data. The results also point towards the increasing utility of bioacoustic foundation models for biodiversity monitoring and real world applications.

XI. FUTURE WORK

Several directions remain open for improving performance on this task. Semi-supervised learning on unlabeled sound-

scapes represents perhaps the most promising avenue: the full `train_soundscapes/` directory contains far more recordings than the 66 files with expert annotations. Pseudo-labeling using the current best model, followed by selective re-training on high-confidence predictions, could substantially increase the volume of domain-adapted training data without requiring additional human annotation.

Larger architectures were not explored due to compute constraints. EfficientNet-B4 or B5 variants, as well as larger AST configurations, may capture more subtle vocalization patterns, particularly for rare species where the model currently struggles most. Test-time augmentation—applying horizontal flipping, time shifting, and frequency shifting to input spectrograms at inference time and averaging multiple predictions per window—could reduce prediction variance at no additional training cost.

Per-class threshold optimization on the soundscape validation set would allow thresholds to be independently tuned for each species, potentially recovering recall for rare species that are currently missed under a global threshold. Finally, multi-resolution spectrogram inputs using multiple FFT window sizes would allow the model to simultaneously capture both short transient calls and longer tonal vocalizations, improving detection across the diverse acoustic repertoires of the 234 target species.

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